```
<!DOCTYPE html>
<html><head>
<script src="//cdnjs.cloudflare.com/ajax/libs/jquery/2.1.1/jquery.min.js"></script>
<script src="//cdnjs.cloudflare.com/ajax/libs/underscore.js/1.7.0/underscore-min.js"></script>
<script src="//cdnjs.cloudflare.com/ajax/libs/backbone.js/1.1.2/backbone-min.js"></script>
<title>Backbone.js playground</title>
</head><body></body></html>
```

You can play with this code online on JS Bin at http://jsbin.com/fineviyejo/1/edit

Yeoman is a great tool for bootstrapping web projects. It properly structures your code in folders, downloads required libraries and builds and packages web apps from the command line.

Backbone.View

Let's start with the bits of Backbone that actually shows something on the page. Here is a bare minimum View that when rendered and added to the page will display a button called "Hello World".

```
var Hello = Backbone.View.extend({
 tagName: 'div',
 className: 'hello',
 render: function() {
  this.$el.html('<button>Hello World</button>');
  return this;  } });
```

Let's unpack this. In this snippet, we are creating a new custom View by extending the base Backbone View. The tagName and className properties define the tag that surrounds our view, and its class. In fact, if we just want a div then we need not even include a tagName since it defaults to div.

The render method is responsible for actually creating the content of the view. Here, it is adding the button labelled 'Hello World'. Backbone views have an 'el' property that represents the raw JavaScript DOM representation of our view. If you are using jQuery then $el will be the jQuery-enhanced version the view.

The Backbone-Eye add-on for FireBug and Firefox is a great way to inspect running Backbone apps.

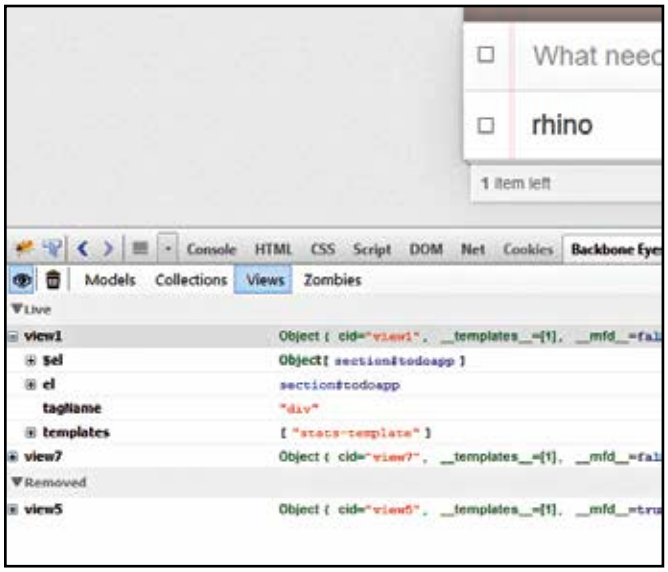
In the render method, we use the jQuery html method to set the content of our view's element. You could have directly manipulated the raw non-jQuery representation via el if you so desired.

To actually have this show up on the page, we first need to create an instance of the above view, and add it to the page body. The following lines of code should take care of that:

```
var h = new Hello;
h.render();
$("body").html(h.$el);
```

We first create a new instance of our new View, we then render it, and finally we set the HTML content of the page body using jQuery to our rendered view accessible via the $el (or el) property.

Backbone is flexible enough that you could possibly automatically render the view in its initialize method and even add it to the page body. This is what the flexibility of Backbone gets you.



View sections under the View tab

Now let's take this a step forward. We want this button to display a counter that shows how many times it has been clicked. The first step in this process will be to add an initialize method that sets this count to "0" (zero) in the beginning. Then we need some way to listen for, and respond to a mouse click event.

Thankfully, Backbone makes both very simple. You can add an attribute called events to the button that maps events to their corresponding methods. Here is our completed code (we removed some unnecessary stuff):

```
var Hello = Backbone.View.extend({
 initialize: function() {
  this.count = 0;
 },
 events: {
  'click button': 'updateCount'
 },
 updateCount: function(event) {
  this.count++;
  this.render();
 },
 render: function() {
  this.$el.html('<button>Hello World: ' + this.count + '</button>');
 } });
```

What's going on in the initialize method should be fairly obvious here; the more interesting bit is in events, the format of which is quite simple: 'event selector': 'callback'

So in the above code we are listening to the click event on the element that matches the selector button and then calling the `updateCount` method in response. You can also listen to other events such as `mouseover`, `dblclick`, or `contextmenu`. The selector bit is the kind you use with jQuery to match elements, so it could be something like #btn.test.example. As you click the button you will see that the view will automatically update on page!

Backbone.Model

Backbone models are as flexible as the Views, leaving a lot to the developer. The most basic model you can create is essentially an